

Accessible Machine Learning Classification of Triple-Negative Breast Cancer Molecular Subtypes

Using Routine Clinical Data

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The Problem: Precision Medicine Gap

15%

of breast cancers
are TNBC

4

molecular subtypes
with different treatments

85%

don't get
subtype-specific care

Why? RNA sequencing costs \$2,000+ and takes 2-3 weeks

Health disparities: Higher incidence in Black women, younger patients

Aggressive biology: Worse prognosis than other breast cancers

Limited options: No targeted therapies for ER-, PR-, HER2- tumors

Four TNBC Molecular Subtypes

Each has different biology and drug sensitivities

BLIS

Basal-Like Immune Suppressed

Biology: High proliferation, DNA damage response

Treatment: PARP inhibitors, platinum chemo

BLIA

Basal-Like Immune Activated

Biology: High immune infiltration, TILs

Treatment: Immunotherapy (checkpoint inhibitors)

LAR

Luminal Androgen Receptor

Biology: AR expression, hormone signaling

Treatment: AR antagonists (from prostate cancer)

MES

Mesenchymal

Biology: Stem-like, EMT features

Treatment: PI3K/mTOR inhibitors

Research Question

Can we accurately classify TNBC subtypes using routine clinical data instead of expensive RNA sequencing?

Current: RNA Sequencing

- Cost: \$2,000 - \$3,000
- Time: 2-3 weeks
- Access: Limited to major centers

Our Approach: Clinical Data

- Cost: \$200 - \$500 (IHC panel)
- Time: Days
- Access: Any pathology lab

What Clinical Data Can We Use?

Demographics

- Age
- Race/ethnicity
- BMI
- Menopausal status

Tumor Pathology

- Tumor size/grade
- Histologic type
- Lymph node status
- TILs score

IHC Markers

- AR (androgen receptor)
- Ki-67 (proliferation)
- CK5/6, EGFR (basal)
- PD-L1, CD8 (immune)

Genetic/Labs

- BRCA1/2, TP53
- Neutrophil/lymphocyte ratio
- LDH
- Routine CBC

All routinely collected - no RNA sequencing.

Methodology

1

Data Collection

TCGA-BRCA (~150 TNBC samples) for training
METABRIC for external validation

2

Feature Engineering

Extract demographics, pathology, IHC, genetics
Handle missing data, create interaction terms

3

Model Training

Random Forest, XGBoost, SVM, Neural Networks
5-fold cross-validation, hyperparameter tuning

4

Validation

Test on held-out TCGA data
External validation on METABRIC cohort

5

Panel Design

Feature importance analysis
Identify minimum test panel for $\geq 70\%$ accuracy

Tools & Technologies

- Python 3.9+
- scikit-learn (ML algorithms)
- XGBoost (gradient boosting)
- PyTorch (neural networks)
- pandas (data manipulation)
- matplotlib/seaborn (viz)

Success Metrics

Primary Goal: $\geq 70\%$ Classification Accuracy

Performance Metrics

- Overall accuracy $\geq 70\%$
- Per-subtype F1-scores ≥ 0.65
- Precision & recall for each subtype
- AUC-ROC curves
- Confusion matrices

Validation Strategy

- 5-fold cross-validation
- Held-out TCGA test set
- External validation on METABRIC
- Performance within 5% across datasets

8-Week Project Timeline

1

Data Collection & Preprocessing

2-3

Feature Engineering & Model Development

4-6

Model Validation & Optimization

7

Clinical Panel Design

8

Final Analysis & Report

Expected Impact

4-10x Cost Reduction: \$200-500 vs \$2,000+ RNA-seq

Clinical Access

- Enable subtyping in community oncology
- No specialized genomics lab needed
- Results available in days vs weeks

Health Equity

- Affordable precision medicine
- Reduce disparities in TNBC care
- Accessible to underserved populations

Research Translation

- Foundation for subtype-specific trials
- Identify patients for targeted therapies
- Enable precision medicine at scale

Summary

Problem: 85% of TNBC patients lack access to molecular subtyping

Solution: ML classification using affordable routine clinical data

Goal: $\geq 70\%$ accuracy at $< \$500$ cost vs $\$2,000+$ RNA-seq

Impact: Democratize precision medicine for TNBC patients

Making molecular subtyping accessible to all